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CLIF Meets RL: A Hands-On Workshop on Training Reinforcement Learning Models with Common Longitudinal ICU Format (CLIF) Data

Session Code: TRU08

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Disclosure of Relevant Financial Relationships

The following presenter(s) **have no** relevant financial relationship(s) with ineligible companies to disclose.

- Saki Amagai, BA
- Kaveri Chhikara, MS
- Yikuan Li, PhD

Learning Objectives

At the conclusion of this presentation, the learner will be able to:

- Describe the **CLIF** data standard and consortium, and use **tools** such as *clifpy*, CLIF Lighthouse, and MIMIC-CLIF to validate and work with standardized ICU data.
- Explain how CLIF's standardized schema and federated analysis approach enable reproducible multi-site **epidemiological** research in critical care, using a **multi-center** ICU readmission study as a model.
- Formulate an ICU sequential decision-making problem as a **Markov Decision Process** and train and evaluate a Double Deep Q-Network on CLIF-formatted trajectories using offline reinforcement learning (**RL**) methods.

CLIF

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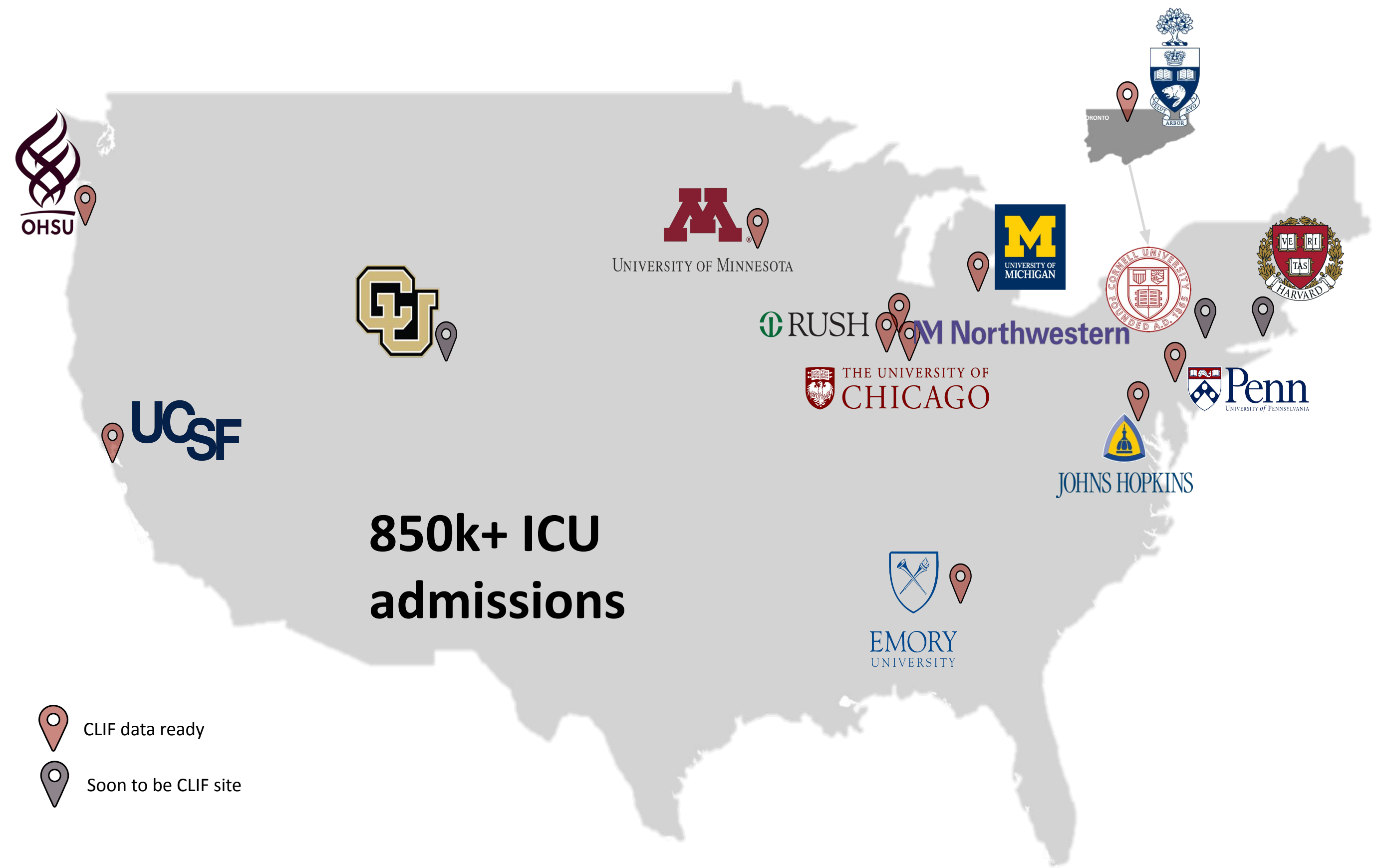

- **Open-source** research format for longitudinal ICU data science
- **Federated** research network- code is shared, not patient data
- Focused on a limited set of **minimum Common ICU data Elements**
- Accessible via **MIMIC-CLIF**

Team Page


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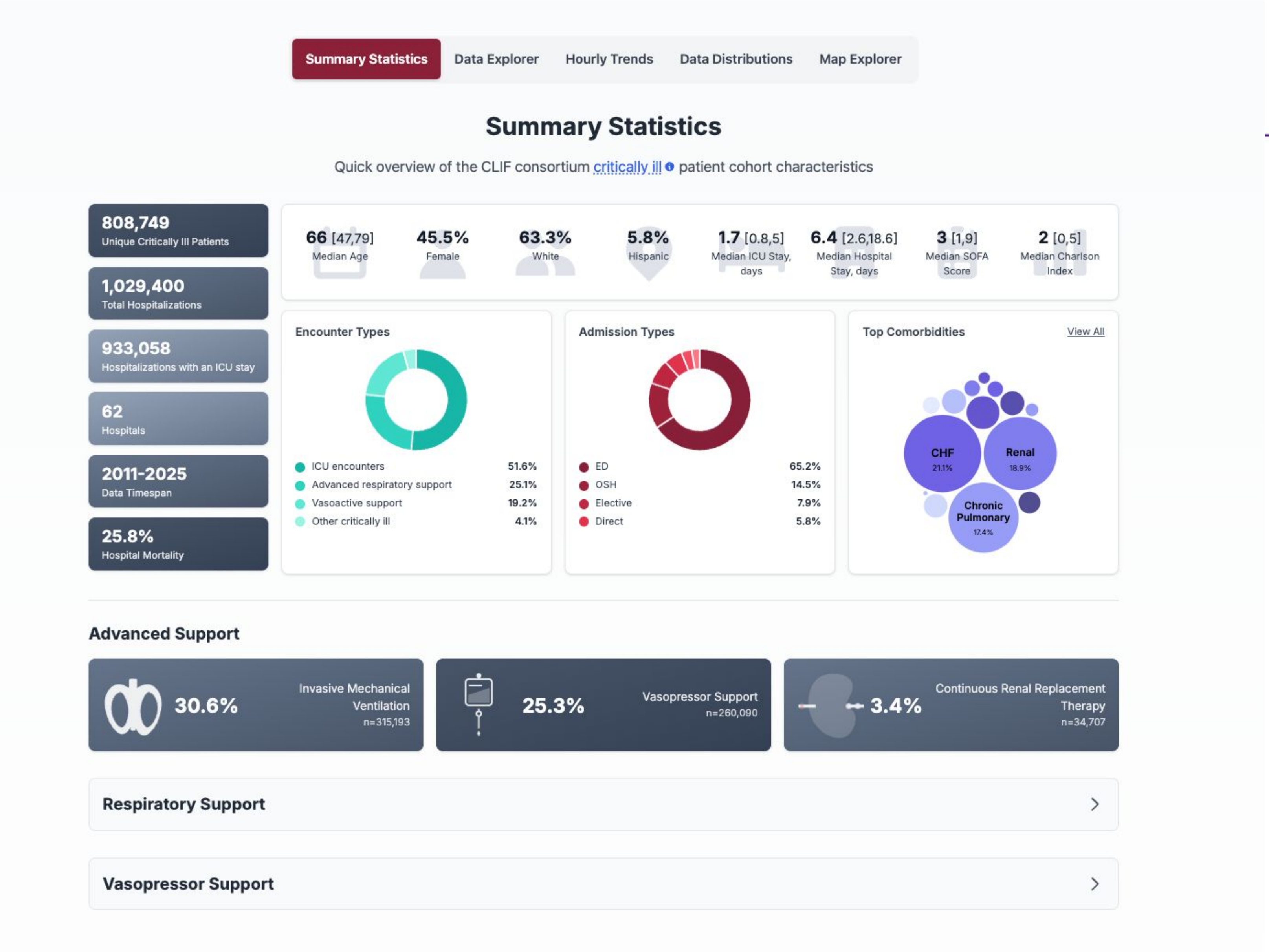
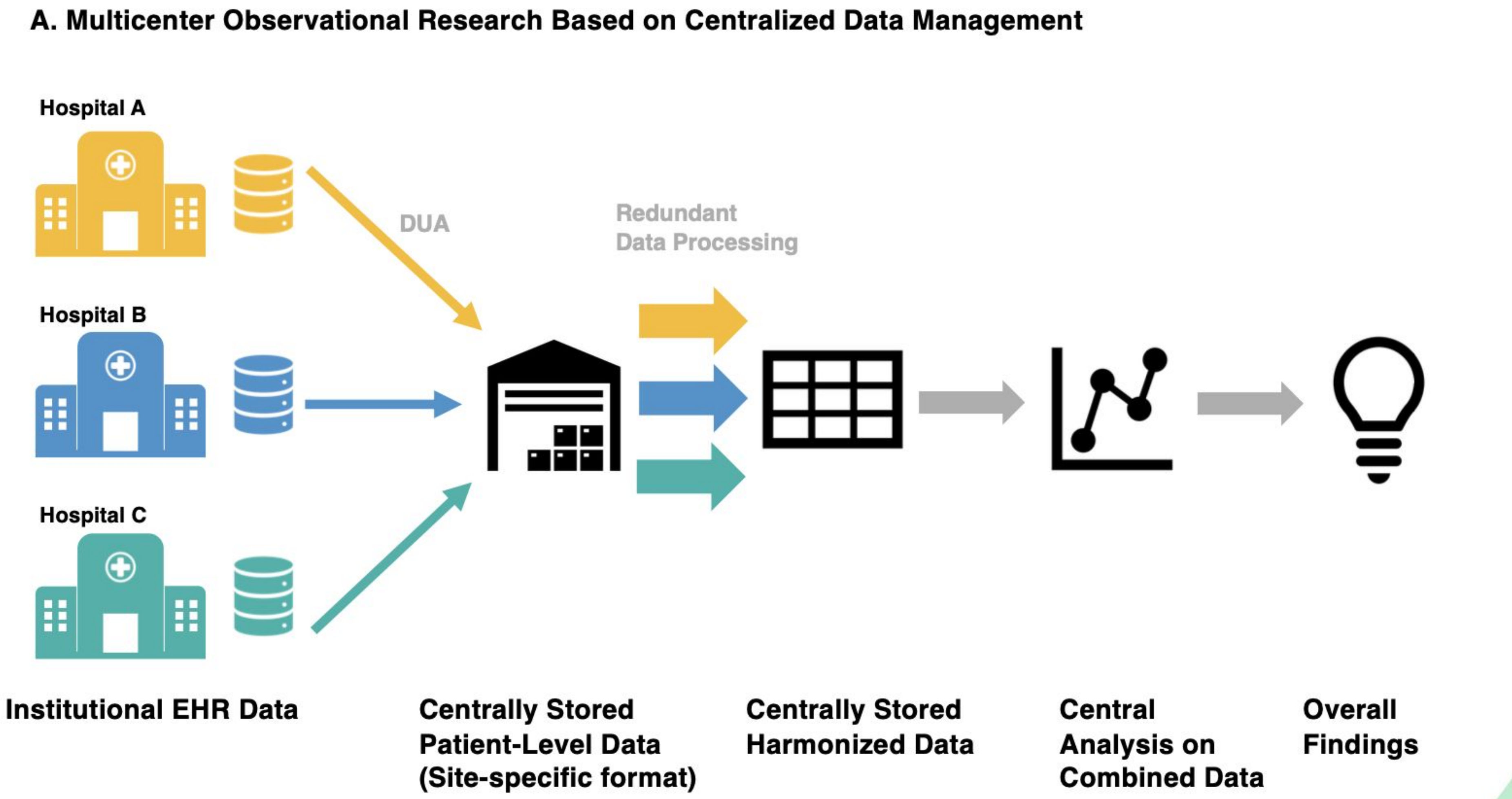
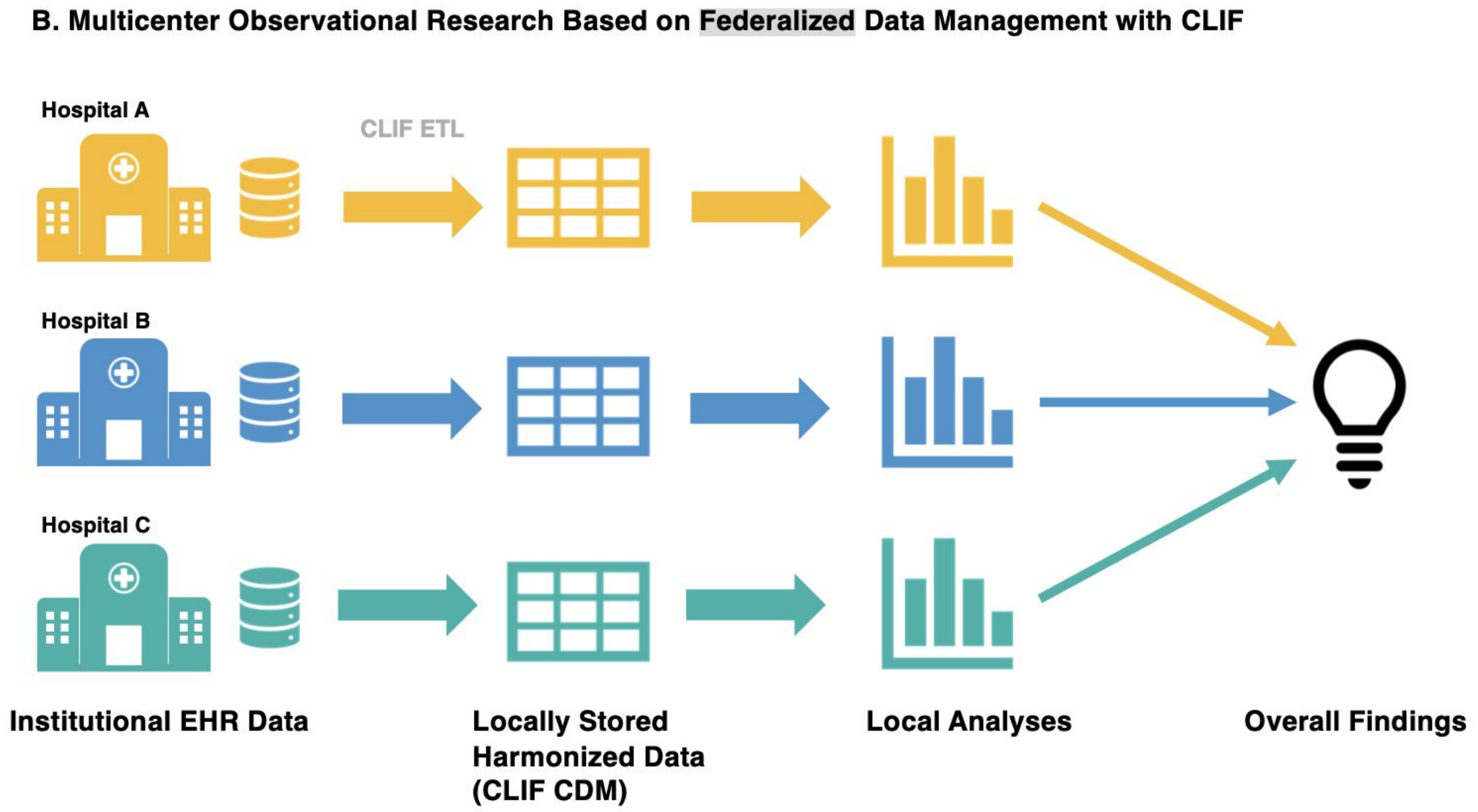
CLIF cohort dashboard



Traditional observational research approach



CLIF Federated research approach



Lead Site

1 Define Project Scope
Develop Code on CLIF database
Validate on MIMIC

UChicago Medicine

5 Analyze results from all sites
Perform meta-analysis

2 Distribute code via GitHub

4 Share aggregate results

Patient-level data is not shared at any step of this process

Participating Sites

Northwestern Medicine

Medical School, University of Minnesota, Driven to Discover

Johns Hopkins Medicine

University of Michigan

Rush

Penn, University of Pennsylvania

Oregon Health & Science University (OHSU)

3 Run analysis scripts on respective CLIF databases

Mission, Vision & Values

Mission

Promote transparency, interoperability, and innovation in critical care data science by developing a consistent and comprehensive ecosystem of essential data elements that describe critical illness.

Vision

CLIF will be adopted worldwide as a research and operational database format, facilitating significant advances in critical care medicine.

Core Values

clif-icu.com/mission

Open Collaboration & Access

Welcoming diverse researchers, clinicians, and developers to strengthen CLIF's ecosystem

Data Privacy & Security

Patient confidentiality is paramount; federated analysis ensures data never transfers between sites

Scientific Integrity

Rigorous research standards with mCIDE for credibility and reproducibility

Transparency & Reproducibility

Code follows FAIR principles while maintaining proprietary data protections

Encouraging Partnerships

Open-source tools under permissive licenses to support worldwide adoption

CLIF DQA Framework => CLIF Quality report

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3 Dimensions - 20 RULES

Over 2000+ checks for 2.1 Beta Tables

Conformance

7 RULES

Structural: Table presence, column presence, category-group mapping

Value: Data types, field formats, mCIDE vocabulary, standardized reference units

VALIDATION RULES

1	Table Presence	STRUCTURAL
2	Column Presence	STRUCTURAL
3	Category Mapping	STRUCTURAL
4	Data Type	VALUE
5	Field Format	VALUE
6	Vocabulary Conformance	VALUE
7	Standardized Units (Labs)	VALUE

Completeness

5 RULES

Atemporal: Column missingness, conditional requirements

mCIDE: Standardized vocabulary coverage

Relational: Foreign key integrity, cross-table presence

VALIDATION RULES

1	Column Missingness	ATEMPORAL
2	Conditional Requirements	ATEMPORAL
3	Vocabulary Coverage	MCIDE
4	Foreign Key Integrity	RELATIONAL
5	Conditional Presence	RELATIONAL

Plausibility

8 RULES

Atemporal: chronology ordering, numeric range, field logic, dose-unit

Temporal: Overlapping periods, category consistency

Uniqueness: Duplicate composite keys

Relational: Cross-table temporal window

VALIDATION RULES

1	Chronological Field Order	ATEMPORAL
2	Numeric Plausibility	ATEMPORAL
3	Field Plausibility	ATEMPORAL
4	Dose-Unit Consistency	ATEMPORAL
5	Temporal Consistency	TEMPORAL
6	Overlapping Event	TEMPORAL
7	Unique Composite Keys	UNIQUENESS
8	Temporal Window	RELATIONAL

Adapted from Kahn et al. (2016) | Implemented in CLIFpy

Conformance (6)

metric	rule	rule_description	column_field	severity	finding
conformance	C.1	Table presence verification	NA	INFO	Table 'adt' present with 964 rows and 10 columns
conformance	C.2	Required columns presence check	__index_level_0__, patient_id	WARNING	Found 2 extra columns not in schema
conformance	C.2	Required columns presence check	NA	INFO	All required columns present
conformance	C.3	Column data type validation	NA	INFO	All column data types match schema
conformance	C.4	Datetime format validation	NA	INFO	All datetime columns are properly formatted
conformance	C.5	Categorical values conformance	location_type	WARNING	Invalid: 'cvicu_icu' (31 rows)

Completeness (5)

metric	rule	rule_description	column_field	severity	finding
completeness	K.1	Required column missingness	NA	INFO	All required columns below thresholds (error≥50.0%, warn≥10.0%): hospitalization_id=0.0%, hospital_id=0.0%, hospital_type=0.0%, in_dttm=0.0%, out_dttm=0.0%, location_category=0.0%
completeness	K.2	Conditional field requirements	location_type	INFO	Conditional requirement satisfied: ICU locations must have location_type specified — 178/178 rows present (100%)
completeness	K.3	mCIDE value coverage	hospital_type	ERROR	Missing 2 mCIDE values: community, LTACH
completeness	K.3	mCIDE value coverage	location_category	ERROR	Missing 5 mCIDE values: l&d, hospice, rehab, radiology, dialysis
completeness	K.3	mCIDE value coverage	location_type	ERROR	Missing 5 mCIDE values: cardiothoracic_surgical_icu, mixed_cardiothoracic_icu, burn_icu, neuro_icu, neurosurgical_icu

Plausibility (19)

metric	rule	rule_description	column_field	severity	finding
plausibility	P.1	Chronological ordering constraints	in_dttm, out_dttm	INFO	Chronological ordering satisfied: ADT in_dttm must be strictly before out_dttm — 964/964 rows valid (100%)
plausibility	P.3	Field-level plausibility rules	location_type	INFO	Field plausibility satisfied: Non-ICU locations should not have ICU location_type — 786/786 rows valid (100%)
plausibility	P.5	Overlapping time period detection	in_dttm, out_dttm	INFO	No overlapping time periods detected for hospitalization_id on in_dttm/out_dttm — 964 records checked
plausibility	P.6	Category temporal consistency	location_category	WARNING	location_category: psych absent in 77/80 years (2110-2201) 2110 2201
plausibility	P.6	Category temporal consistency	location_category	WARNING	location_category: other absent in 49/80 years (2110-2201) 2110 2201

Sample ADT Quality Report on CLIF Demo data

Common ICU data science code => CLIFpy

CLIF

Transform critical care data into actionable insights

pypi v0.4.4 | python 3.9 | License Apache 2.0 | docs latest | uv

Documentation | Quick Start | CLIF Website

Standardized framework for critical care data analysis and research

CLIFpy is the official Python implementation for working with CLIF (Common Longitudinal ICU data Format) data. Transform heterogeneous ICU data into standardized, analysis-ready datasets with built-in validation, clinical calculations, and powerful data manipulation tools.

Key Features

- 📊 **Comprehensive CLIF Support**: Full implementation of all CLIF 2.0 tables with automatic schema validation
- 🏥 **Clinical Calculations**: Built-in SOFA scores, comorbidity indices, and other ICU-specific metrics
- 💊 **Smart Unit Conversion**: Automatically standardize medication dosages across different unit systems
- 🔗 **Encounter Stitching**: Link related ICU stays within configurable time windows
- ⚡ **High Performance**: Leverages DuckDB and Polars for efficient processing of large datasets
- 🌍 **Timezone Aware**: Proper timestamp handling across different healthcare systems
- 📈 **Wide Format Support**: Transform longitudinal data into hourly resolution for analysis

clifpy 0.4.4

```
pip install clifpy
```

Feature	Coder	Clinician	Status
SOFA Score Computation	Kaveri Chhikara	Will Parker	Published
Respiratory Support Waterfall	Kaveri Chhikara	Nick Ingraham	Published
Presumed Infection and ASE	Vaishvik Chaudhari	Kevin Buell	Published
Cumulative Dose Calculation	Zewei (Whiskey) Liao		Future Feature
Lab Conversion Units	Dema Therese	Cathy Gao	In Progress
CRRT Waterfall	Kaveri Chhikara	Shan Guleria, Jay Koyner	Future Feature
Medication Unit Conversion	Zewei (Whiskey) Liao		Published

Anyone can code a CLIF project via CLIF-MIMIC

PhysioNet

Share About FAQs Explore

Database | Credentialed Access

MIMIC-IV-Ext-CLIF: MIMIC-IV in the Common Longitudinal ICU data Format (CLIF)

Zewei Liao , Shan Guleria , Kevin Smith , Rachel Baccile , Kaveri Chhikara , Dema Therese , Vaishvik Chaudhari , Michael Craig Burkhart , Brett Beaulieu-Jones , Snigdha Jain , Kathryn Connell , Kevin Buell , Juan Rojas , Patrick Lyons , Siva Bhavani , Catherine A Gao , Chad Hochberg , Nick Ingraham , William Parker , CLIF Consortium

Published: March 23, 2026. Version: 1.1.0

When using this resource, please cite: Cite | Copy BibTeX

Liao, Z., Guleria, S., Smith, K., Baccile, R., Chhikara, K., Therese, D., Chaudhari, V., Burkhart, M. C., Beaulieu-Jones, B., Jain, S., Connell, K., Buell, K., Rojas, J., Lyons, P., Bhavani, S., Gao, C. A., Hochberg, C., Ingraham, N., Parker, W., & Consortium, C. (2026). MIMIC-IV-Ext-CLIF: MIMIC-IV in the Common Longitudinal ICU data Format (CLIF) (version 1.1.0). *PhysioNet*. RRID:SCR_007345. https://doi.org/10.13026/j481-g420

Please include the standard citation for PhysioNet: (show more options)

Goldberger, A., Amaral, L., Glass, L., Hausdorff, J., Ivanov, P. C., Mark, R., ... & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation [Online]. 101 (23), pp. e215–e220. RRID:SCR_007345.

Contents

Parent Projects

MIMIC-IV-Ext-CLIF: MIMIC-IV in the Common Longitudinal ICU data Format (CLIF) was derived from:

- MIMIC-IV v3.1

Please cite them when using this project.

Share

Access

Access Policy:
Only credentialed users who sign the DUA can access the files.

Abstract

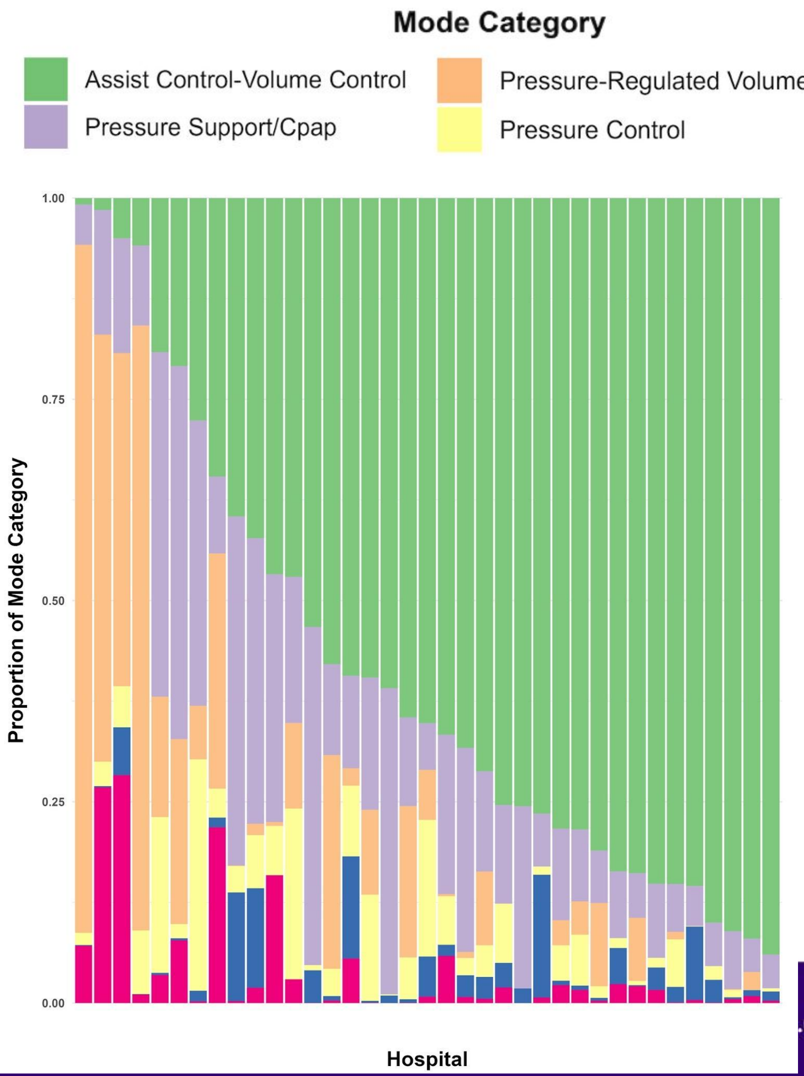
MIMIC-IV-Ext-CLIF is a derived dataset of MIMIC-IV v3.1, transformed into the Common Longitudinal ICU data Format (CLIF). CLIF is an open-source critical care data model developed by a consortium of more than 10 US academic medical centers to standardize intensive care unit (ICU) data for multi-center research. While CLIF has demonstrated value in federated research settings, access has been limited to consortium members with institutional electronic health record (EHR) data. This dataset addresses that gap by providing CLIF-formatted data derived from the freely accessible de-identified MIMIC-IV dataset, enabling researchers worldwide to utilize the CLIF format without requiring institutional EHR access.

MIMIC-IV-Ext-CLIF contains 14 CLIF tables as of the latest CLIF 2.1.0 version, covering core demographics (patient, hospitalization), clinical monitoring (vitals, labs), respiratory support, medication administration, and specialized ICU interventions (e.g., continuous renal replacement therapy). The transformation employs a

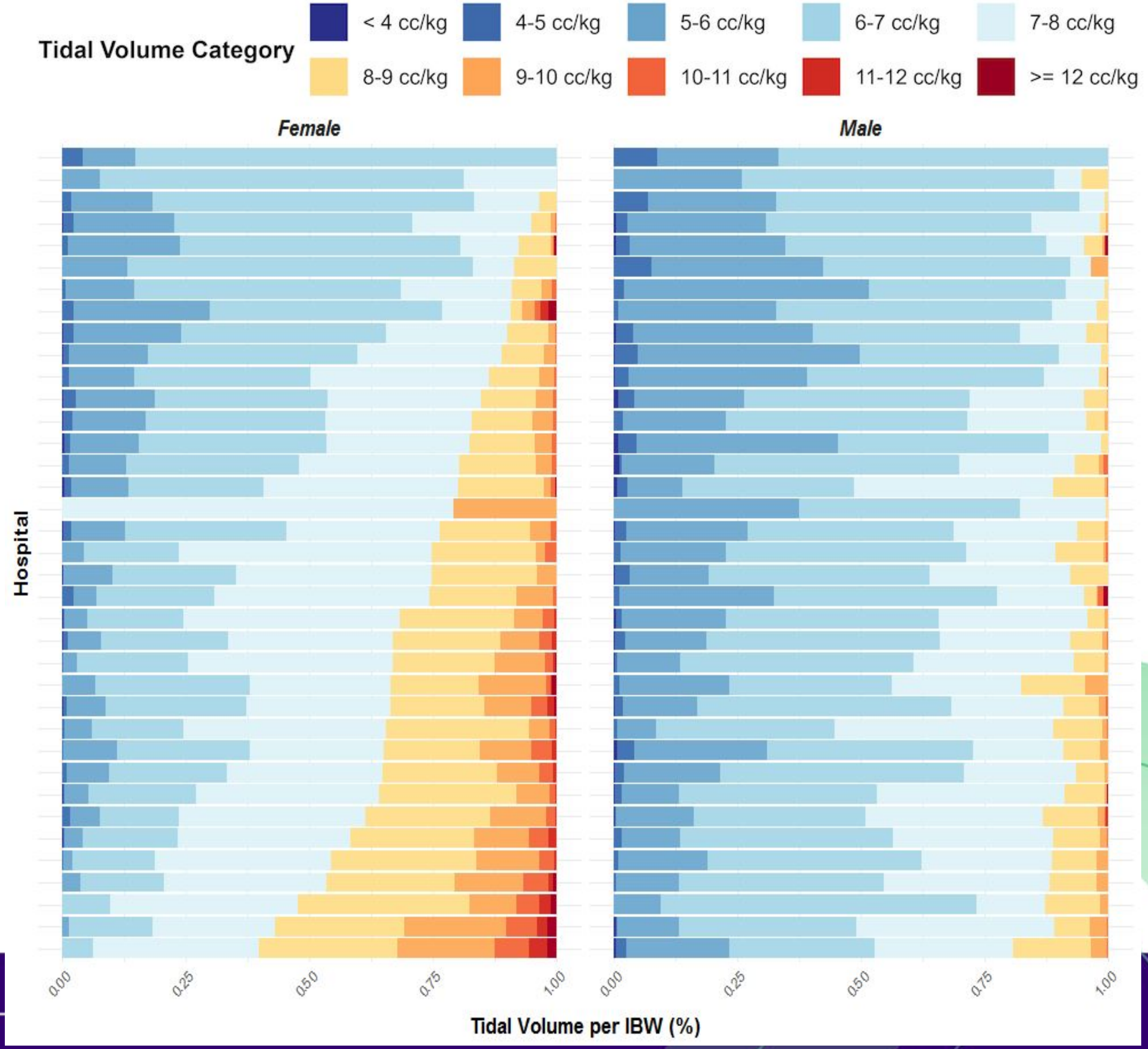
Tools

- Clifpy
- CLIF Quality Report and dashboard
- mCIDE explorer
- CLIF GitHub Organization
- CLIF AI setup:
  - CLIF Skills for Claude
  - CLIF Claude setup
  - CLIF Skills advanced
- CLIF-101 tutorials

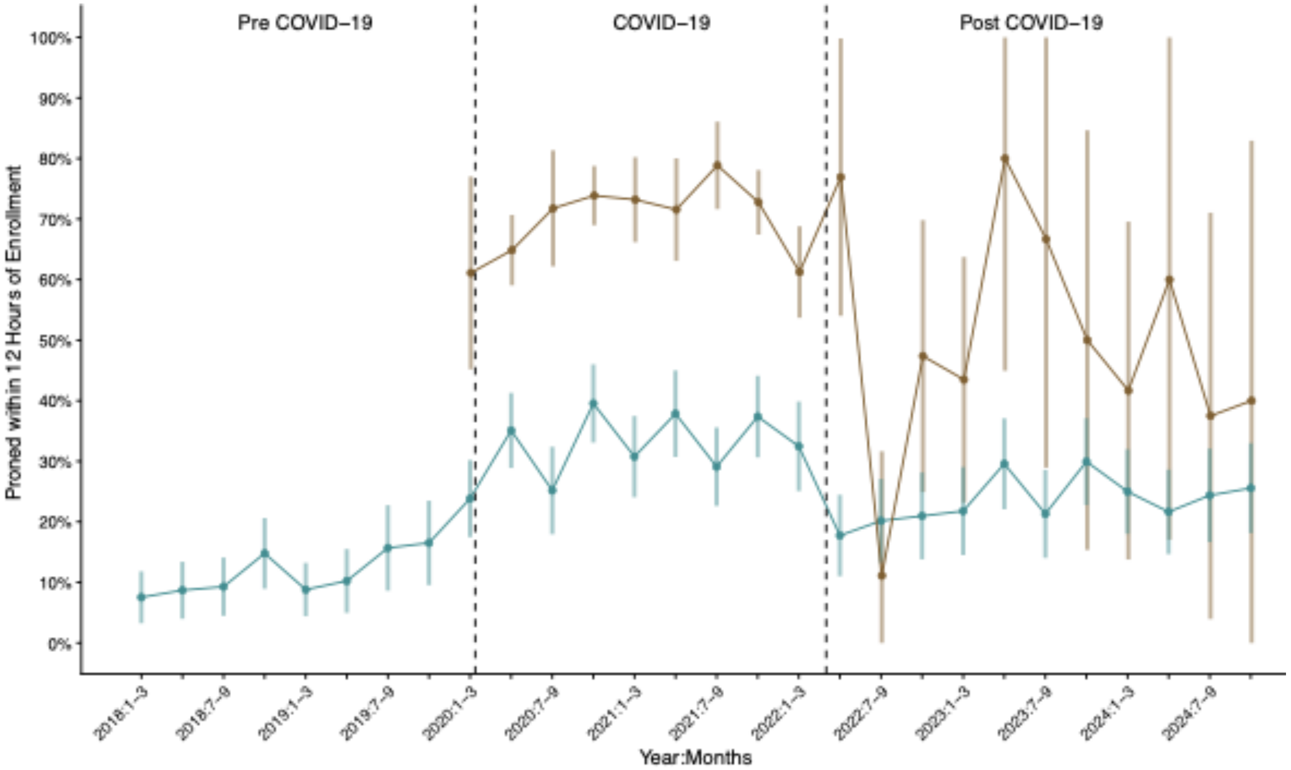
CLIF in Action: Variation in ventilation



Nick Ingraham, MD
University of Minnesota



CLIF in Action: Prone position ventilation over time



Chad Hochberg, MD

Anna Barker, MD, PhD

In press, CCM

CLIF in Action: Lets do it!

- Hypothesis
  - What?
- Cohort Identification
  - Who?
- Methodology
  - How?

CLIF in Action: Lets do it!

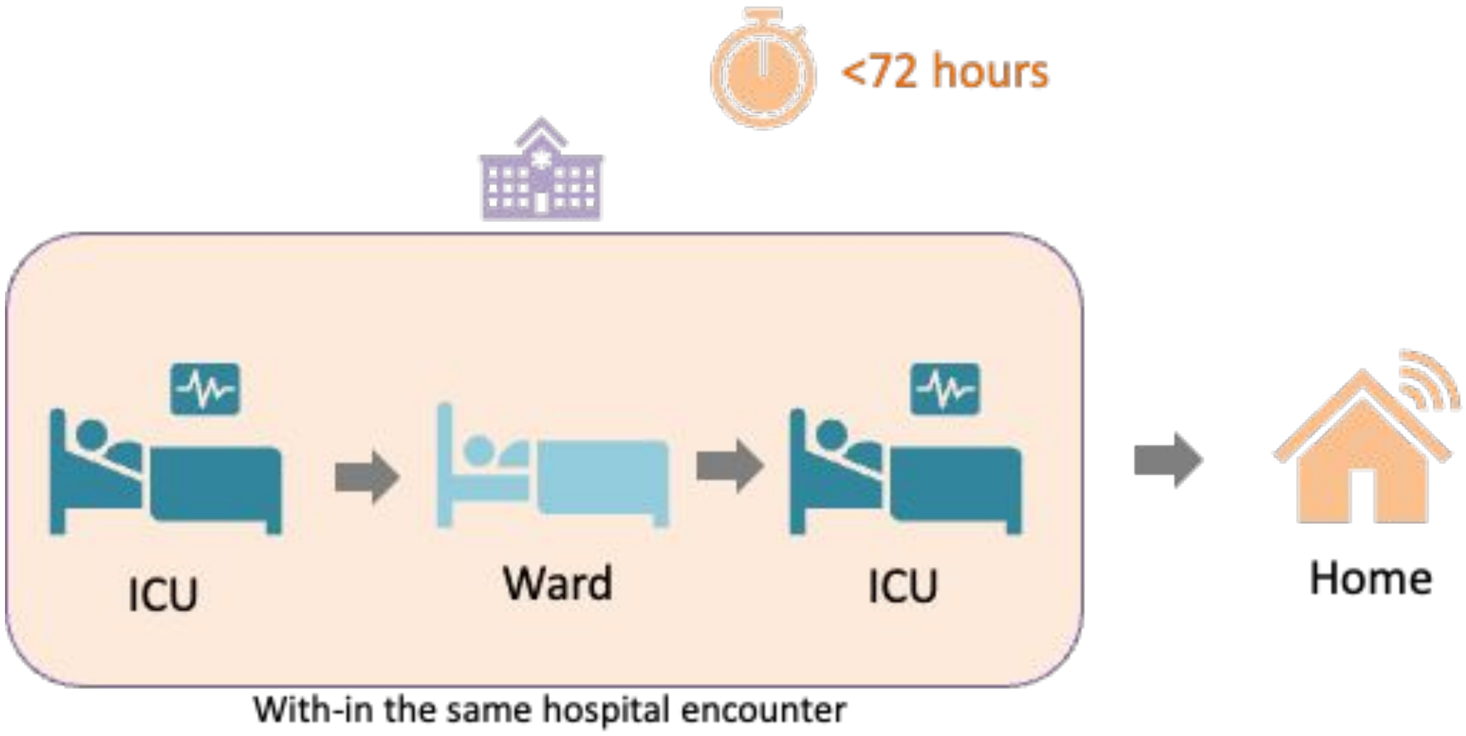
- Hypothesis
  - What?
- Cohort Identification
  - Who?
- Methodology
  - How?

CLIF in Action: Deep dive with Saki

The intensive care unit (ICU) is where the **sickest** patient stays

Over the past two decades, **ICU mortality** in the U.S. has **declined** significantly… but many of them subsequently **bounce back to the ICU**.

ICU Readmission



Why are readmissions bad?

Harmful for patients

- Readmissions are associated with **higher mortality** and **worse health outcomes**

Costly for Health Systems and Patients

- Readmissions are **expensive** (billions in avoidable spending each year)
- **CMS penalizes** hospitals under Hospital Readmission Reduction Program (**HRRP**)

Incomplete care or poor quality

- **Failure** in **care transitions**, premature **discharge**, inadequate **discharge planning**

Critical Care Explorations
SCCM JOURNALS

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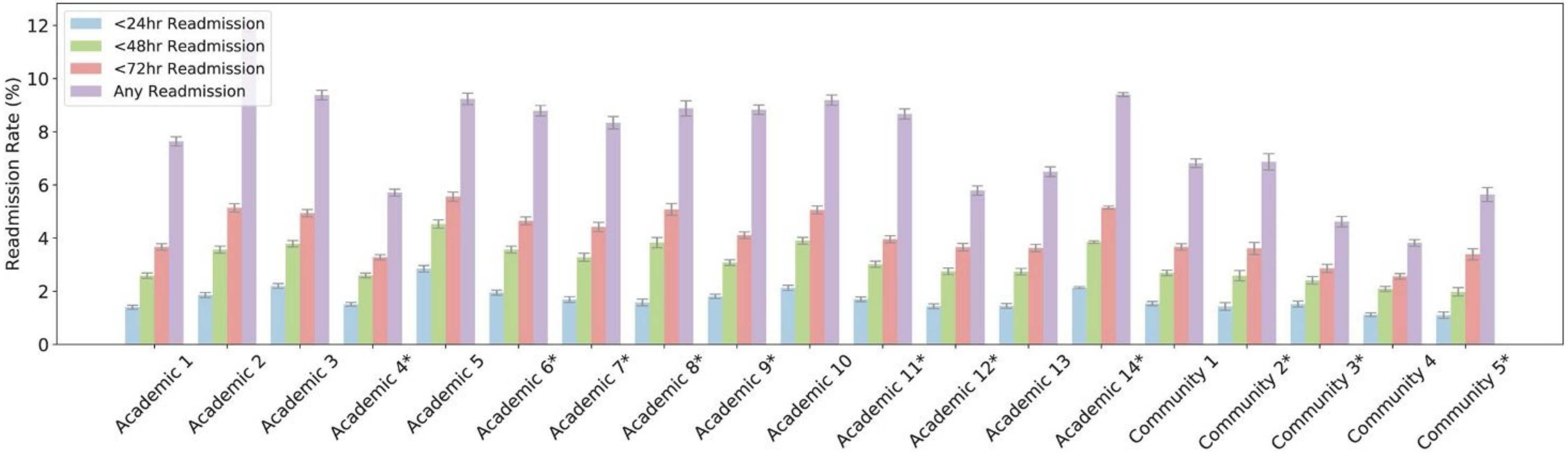
LETTER TO THE EDITOR

The Epidemiology of ICU Readmissions Across Ten Health Systems

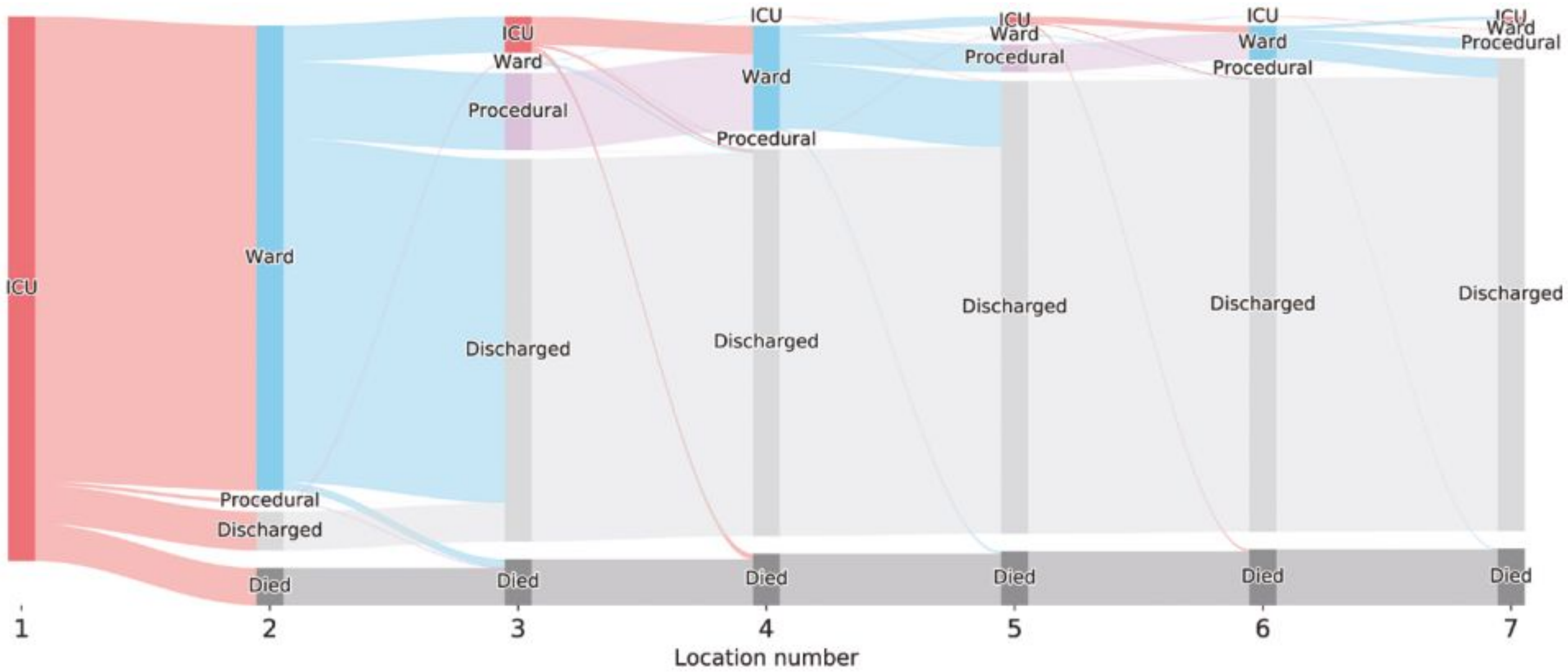
Amagai, Saki BA[1]; Chaudhari, Vaishvik MS[2]; Chhikara, Kaveri MS[3]; Ingraham, Nicholas E. MD[4]; Hochberg, Chad H. MD[5]; Barker, Anna K. MD[6]; Mao, Chengsheng PhD[1]; Ortiz, Alexander C. MD[7]; Weissman, Gary E. MD[8]; Schmid, Benjamin E. MS[8]; Schwinne, Megan MPH[9]; Bhavani, Sivasubramanium V. MD[10]; Guleria, Shan MD[3]; Liao, Zewei MS[3]; Markov, Nikolay MS[11]; Lyons, Patrick G. MD[12]; Park-Egan, Brenna MS[12]; Parker, William F. MD[3]; Luo, Yuan PhD[1]; Rojas, Juan C. MD[2]; Gao, Catherine A. MD[11]; The Common Longitudinal ICU data Format (CLIF) Consortium



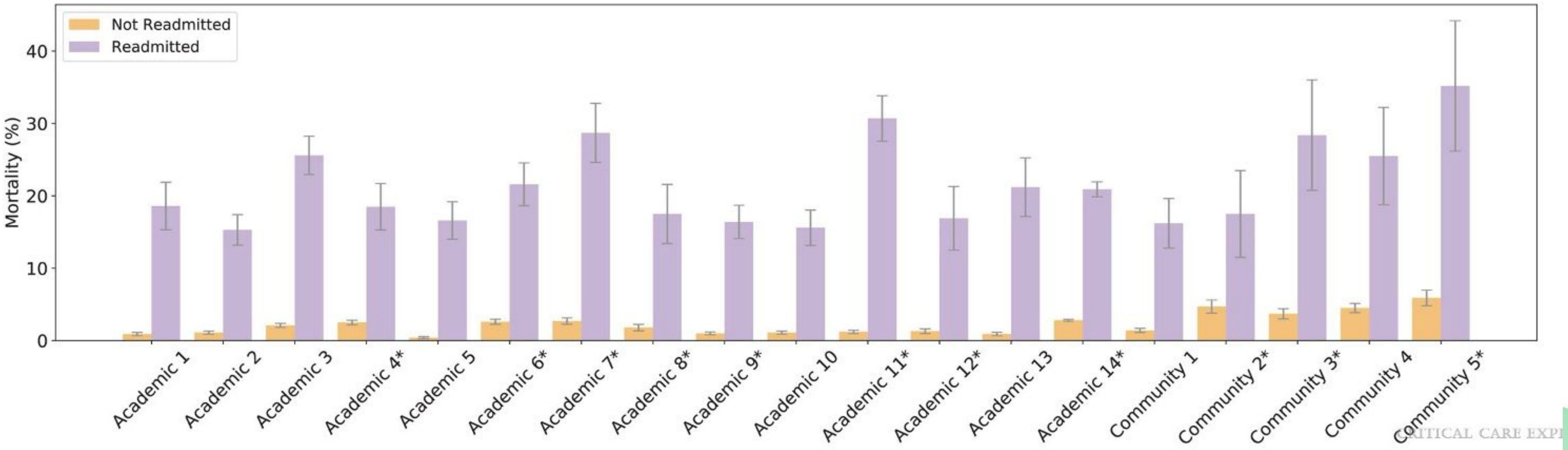
Sample Cohort Characteristics

Feature	Not Readmitted ($n = 169{,}395$)	Readmitted ($n = 15{,}846$)	p
Age, mean (SD)	61.1 (17.2)	63.1 (15.8)	< 0.001
ICU length of stay, d, mean (SD)	3.6 (5.8)	5.7 (9.8)	< 0.001
ICU read-mission, hr, mean (SD)	NA	123.1 (205.0)	NA
Sex: female, n (%)	75,973 (44.8)	6,756 (42.6)	< 0.001
Race: Black, n (%)	40,709 (24.0)	4,086 (25.3)	< 0.001
Race: White, n (%)	103,882 (61.3)	9,395 (59.3)	
Race: Asian, n (%)	5,760 (3.4)	547 (3.5)	
Race: other/ unknown, n (%)	19,044 (11.2)	1,818 (11.5)	
Ethnicity: Hispanic, n (%)	9,263 (5.5)	815 (5.1)	< 0.001
In-hospital mortality, n (%)	3,593 (2.1)	3,257 (20.6)	< 0.001

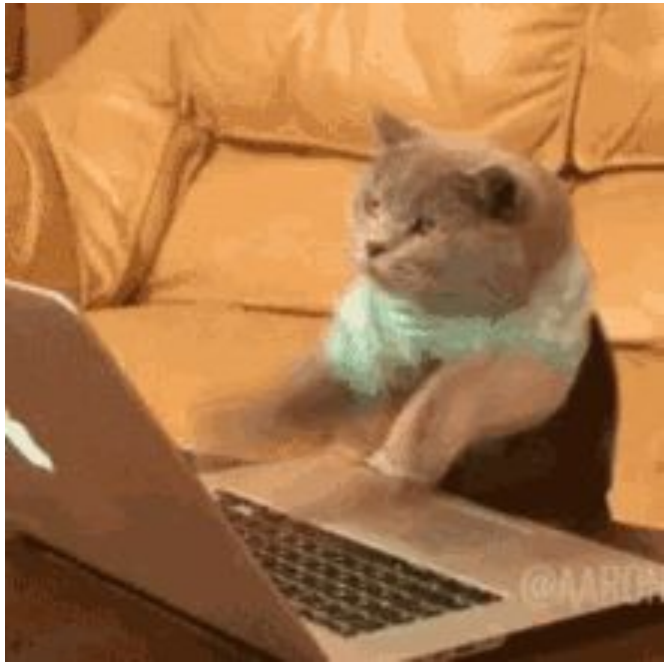
Example Sankey diagram from academic hospital 3 illustrating the patient journey within the hospital after index ICU admission.



Comparison of mortality rates between readmitted and nonreadmitted patients



Now let's try running some code to get the **ICU readmission rates** in our demo dataset!



What is Reinforcement Learning (RL)?

RL is a framework for sequential decision-making. An agent makes a series of decisions over time, where each choice affects what comes next — and it learns from the consequences.

Game-playing (AlphaGo)

The agent picks moves across a full game.

Goal: win the game

Self-driving cars

The car decides to steer, brake, or accelerate — each choice changes the road ahead.

Goal: reach the destination safely

Ventilator management

Clinicians adjust FiO_2, PEEP, and tidal volume over hours to days; each setting shifts the patient's trajectory.

Goal: save the patient

Cookbook for SL and RL

Every ML paradigm follows a recipe: ingredients → model → training → evaluation. RL keeps the structure but swaps the ingredients.

Supervised learning

Ingredients
input x, outcome y

Model
learn f so that $f(x) = \hat{y}$

Training
shrink the gap between $\hat{y}$ and y

Evaluation
check $\hat{y}$ vs. y on a held-out set

Reinforcement learning

Ingredients
a Markov Decision Process (MDP)

Model
learn a *policy* — what action to take in each state

Training
improve the policy so it earns more reward over time

Evaluation
estimate how well the policy would perform if deployed

Ingredient - Markov Decision Process (MDP)

An MDP is the formal recipe for sequential decision-making — five ingredients, illustrated through a self-driving car.

State (S) what's going on now	Speed, position on the road, distance to other cars, traffic light color.
Action (A) what the agent does	Steer left or right, accelerate, brake, change lanes.
Transition (P) how the world reacts	$P(s' \mid s, a)$ — probability of next state s' given state s and action a. Brake on wet road → the car slows down (and may slide a little).
Reward (R) the feedback signal	+ for smooth progress toward destination; − for hard braking, sudden swerves, or collisions.
Discount (γ) how much future matters	Arriving today is worth more than arriving tomorrow — γ controls how patient the agent is.

> Markov property: the next state depends only on the *current* state and action — not the full history. The current state is assumed to capture everything that matters.

Ingredient - Markov Decision Process (MDP) in RL4H

Raghu A, Komorowski M, Ahmed I, Celi L, Szolovits P, Ghassemi M. *Deep Reinforcement Learning for Sepsis Treatment. NeurIPS ML4H Workshop (2017). arXiv:1711.09602.*

Task: learn the optimal IV fluid + vasopressor dosing policy for sepsis using deep RL on MIMIC-III (~17,900 patients). Trajectories built in 4-hour time steps from 24h before to 48h after sepsis onset.

State (S) patient status	47 features (demographics, vitals, labs, fluids/pressors received) kept in *continuous* form — no clustering, fed directly into a neural net.
Action (A) treatment decision	25 discrete actions = 5 IV-fluid dose bins × 5 vasopressor dose bins per 4-hour window.
Transition (P) how the patient evolves	Not modeled. Model-free deep RL — the agent learns Q(s, a) directly from observed trajectories without ever estimating *P(s′ | s, a)*.
Reward (R) clinically shaped	Intermediate: penalize high/increasing SOFA & lactate; reward improvements. Terminal: $\pm R_{max}$ for survival vs. death. Designed by clinicians.
Discount (γ) standard choice	Standard value close to 1 (γ ≈ 0.99) — future rewards barely discounted across the 72-hour horizon.

Model-free v.s. Model-based

The split comes down to one question: do we have a model of the transition $P(s' \mid s, a)$?

Model-based

e.g. self-driving cars, AlphaGo

Transition

We know or can simulate $P(s' \mid s, a)$ — physics for a car, rules of the game for Go.

How the agent learns

Plans ahead by rolling out imagined trajectories through the model, then picks the best action.

Strengths

Sample-efficient and safe to explore in simulation.

Model-free

e.g. sepsis treatment (Raghu et al., 2017)

Transition

No reliable model — patient physiology is too complex and heterogeneous. We skip P entirely.

How the agent learns

Learns the value of state-action pairs directly from observed trajectories — what worked, what didn't.

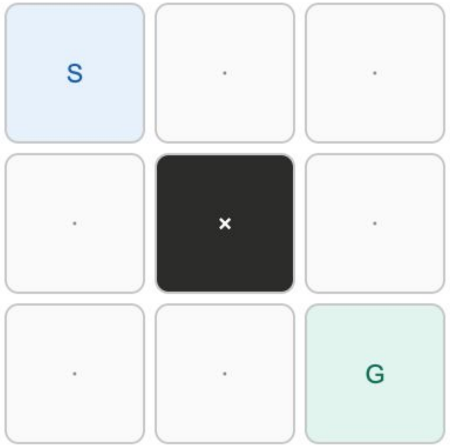
Strengths

No simulator needed. Works on whatever real-world ICU data we already have.

Model - Q Learning

Q-learning learns a table of values *Q(s, a)* — "if I'm in state *s* and take action *a*, how good is that?" — by trial and error.

A tiny grid world



S start · G goal (+1) · × wall
4 actions per cell: up, down, left, right

Q-table size: 9 cells × 4 actions = `36 values`

1. **Start with an empty table**
   All 36 Q-values initialized to 0. The agent knows nothing.

2. **Wander around**
   Pick actions (mostly random at first), watch what reward comes back and where you land next.

3. **Update the table after each step**
   Nudge *Q(s, a)* toward "the reward I just got, plus the best Q-value I can reach from the next state":
   $Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [\, r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a) \,]$
   old estimate + small step toward (reward now + best future value)

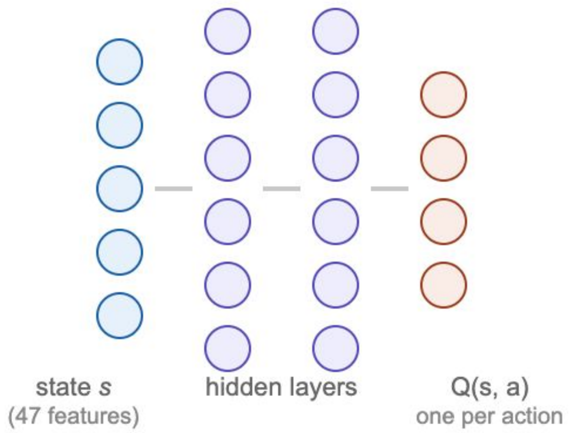
4. **Repeat — values propagate backward**
   The cell next to the goal learns its value first. Then the cell next to that. Eventually every cell knows the best direction to head.

Model - Deep Q Network

Why we can't use a table for the ICU

47 continuous features → essentially infinite states. A Q-*table* would need a row for every patient situation that ever existed. Solution: replace the table with a neural network.

Deep Q-Network



Same idea as Q-learning

The network takes a patient state as input and outputs a Q-value for each possible action. We still want $Q(s, a)$ to predict the long-term reward of taking action a in state s.

Trained with the same target

Minimize the squared error between the network's $Q(s, a)$ and the Bellman target:

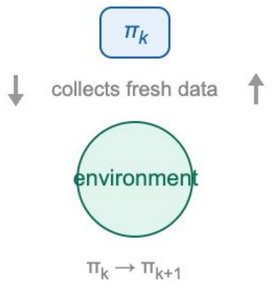
$$loss = (r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a))^2$$

Training - Offline, vs, Off-policy, vs Online

Three flavors of RL differ in when the agent gets data and whose policy collected it.

A

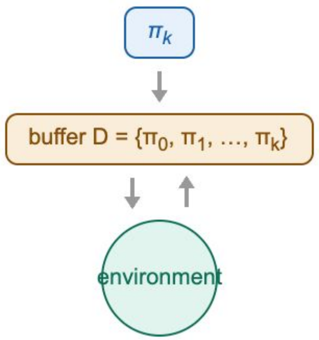
Online (on-policy)



Agent π_k trains on data it just collected itself, then updates to π_{k+1}. Old data is thrown away.

B

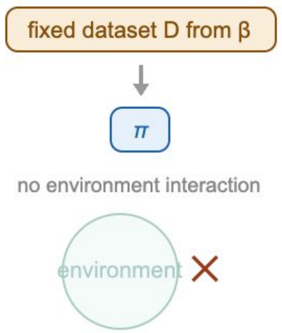
Off-policy



A growing buffer keeps data from all past policies. The agent still interacts with the environment to add fresh data.

C · HEALTHCARE

Offline (batch)



Data was collected once by an unknown behavior policy β (the clinicians). The agent learns from *D* only — no interaction, no exploration — then deploys.

WHY HEALTHCARE RL IS OFFLINE

We can't have an RL agent try random vasopressor doses on real patients to learn. The only data we have is historical ICU records — what clinicians actually did and what happened next. So clinical RL is always offline, and our methods have to work without ever exploring.

Diagram adapted from Levine, Kumar, Tucker, Fu. Offline Reinforcement Learning: Tutorial, Review, and Perspectives on Open Problems. arXiv:2005.01643 (2020), Figure 1.

Evaluation

THE DILEMMA

In games, we can play the new policy and see if it wins. In the ICU, we can't deploy an untested policy on real patients — so how do we know if it's any good?

All we have is a fixed dataset of clinician decisions. Two strategies have emerged for evaluating a learned policy without ever running it:

Off-policy evaluation

a value estimate, in numbers

"If we deployed the learned policy, what return would we expect?"

Use the historical data to **simulate** the new policy's performance, by re-weighting trajectories based on how closely the clinicians' actions matched the AI's recommendations.

Gives you: a single number — the estimated expected return of the learned policy.

Concordance analysis

an empirical mortality plot

"Among patients with similar baseline risk, did clinician agreement with the AI lower mortality?"

Group patients by **how closely the clinician's dose matched the AI's recommendation**, then compare mortality across groups while adjusting for baseline risk factors.

Gives you: an adjusted association between AI-agreement and outcomes — clinically intuitive.

Evaluation (cont'd)

Same data, two complementary evaluations — one statistical, one clinical.

Off-policy evaluation: weighted importance sampling

STEP 0 — LEARN A BEHAVIOR POLICY

Clinicians don't give us probabilities, so we fit a model $\pi_\beta(a \mid s)$ that predicts what the clinician would do in any state — e.g. a classifier on the same MIMIC data.

PATIENT	PER-STEP RATIO $\Pi_E(A\|S) / \Pi_B(A\|S)$	P_I	R_I
τ_1	0.9 × 0.8 × 0.9 × 0.7	0.45	+100
τ_2	0.3 × 0.2 × 0.4 × 0.5	0.01	−100
τ_3	0.8 × 0.7 × 0.9 × 0.8	0.40	+100
τ_4	0.6 × 0.5 × 0.3 × 0.4	0.04	−100

$$\text{WIS} = \Sigma\ \rho_i R_i \ / \ \Sigma\ \rho_i$$

$$= (0.45 \cdot 100 - 0.01 \cdot 100 + 0.40 \cdot 100 - 0.04 \cdot 100)$$

$$/ \ (0.45 + 0.01 + 0.40 + 0.04)$$

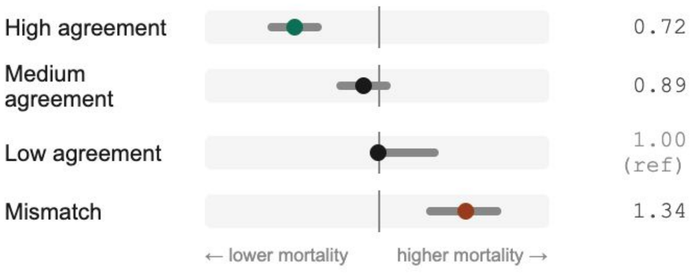
$$= 80 \ / \ 0.90 \rightarrow +89$$

Reading it: patients where the AI agreed with the clinician (high ρ) dominate the estimate. Here, the high-ρ patients all survived, so the estimated AI value is high. **Caveat:** two patients carry ~95% of the weight — the estimate is fragile, with wide confidence intervals.

Concordance: adjusted mortality by AI agreement

"Among patients with similar baseline risk, did clinician agreement with the RL policy lower mortality?"

Adjusted for: age, sex, comorbidities, severity score (SOFA / APACHE), via regression.



Output: adjusted odds ratio of mortality at each level of clinician-AI agreement. Higher agreement → lower mortality, after controlling for baseline risk. **Clinically intuitive, but observational** — patients who happened to be treated like the AI may differ in unmeasured ways.

Now let's try running some code to get the **RL agent** in our demo dataset!